1 – install AD

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```
##Valid Code 10-25-2023
Initialize-Disk -Number 1 -PartitionStyle GPT
New-Partition -DiskNumber 1 -DriveLetter Y -UseMaximumSize
Format-Volume -DriveLetter Y -FileSystem NTFS



Install-WindowsFeature -Name AD-Domain-services, DNS, FS-DFS-NameSpace,FS-DFS-Replication -IncludeAllSubFeature -IncludeManagementTools
##Install-WindowsFeature  -IncludeAllSubFeature -IncludeManagementTools
Get-WindowsFeature | Where-Object {$_. installstate -eq "installed"} | Format-List Name,Installstate | more | out-file c:\temp\features.txt

Import-Module ADDSDeployment
Install-ADDSForest `
-CreateDnsDelegation:$false `
-DatabasePath "C:\Windows\NTDS" `
-DomainMode "WinThreshold" `
-DomainName "Blackcloudnetwork.local" `
-DomainNetbiosName "blackcloudNETWORK" `
-ForestMode "WinThreshold" `
-InstallDns:$true `
-LogPath "Y:\Logs" `
-NoRebootOnCompletion:$true `
-SysvolPath "Y:\SYSVOL" `
-Force:$true

##
##$HashArguments = @{
##   CreateDNSDelegation = $false
##    DatabasePath       = "Y:\NTDS"
##   DomainMode         = '7'
##   DomainName         = "blackcloudnetwork.local"
##   ForestMode         = '7'
```

```
##   LogPath           = "Y:\Logs"
##   SysvolPath        = "Y:\SYSVOL"
##   NoRebootOnCompletion= $true
##}
##Install-ADDSForest @HashArguments
##Need to restart After installing forest



## Configure DNS
Get-DnsServerRootHint | Remove-DnsServerRootHint -Force

$timeSpan = New-TimeSpan -Days 7
##Set-DnsServerScavenging -ScavengingState $true -ScavengingInterval "$timeSpan" -RefreshInterval $timeSpan

Set-DnsServerScavenging -ApplyOnAllZones -ScavengingState $true -ScavengingInterval $timeSpan -RefreshInterval $timeSpan


#create Reverse Lookup Zone
Add-DnsServerPrimaryZone -NetworkID "10.10.10.0/24" -DynamicUpdate Secure -ReplicationScope "Forest"


##Add Subnet For Site and Services
New-ADReplicationSubnet -Name "10.10.10.0/24" -Site Default-First-Site-Name


##Install AD Certificate Services
Get-WindowsFeature -Name AD-Certificate | Install-WindowsFeature



##Create OU Structure
Import-Module activedirectory

##Create OU Structure
```

```
Import-Module activedirectory

##Get Disinguished Name
$domainDN = (Get-ADDomain).DistinguishedName
##Adding Quotes to DN
$domainDNQuotes = "`"$domainDN`""

$orgs = Get-Content C:\it\Book1.csv
foreach ($org in $orgs)
{
New-ADOrganizationalUnit -Name $org -Path $domainDN
}

<#  Old code 3-12-2024
$LOC= "DC=blackcloudnetwork,DC=local"
$orgs = Get-Content C:\it\Book1.csv
foreach ($org in $orgs)
{
New-ADOrganizationalUnit -Name $org -Path $LOC
}
>#
```

3 - GPO linking

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```
# Define the list of GPO name patterns with wildcards at the end
$patterns = @(
   "DOD Adobe*",
   "DOD GOOGLE*",
   "DOD Internet*",
   "DoD Microsoft Defender*",
   "DoD Windows Defender Firewall*",
   "DoD Office 2019-M365*",
   "DoD Microsoft Edge STIG*",
   "DoD Mozilla Firefox STIG*"
)

# Initialize an empty array to store matching GPO names
$matchingGPOsArray = @()

# Retrieve matching GPOs
foreach ($pattern in $patterns) {
   $matchingGPOs = Get-GPO -All | Where-Object { $_.DisplayName -like $pattern }
   if ($matchingGPOs) {
      foreach ($gpo in $matchingGPOs) {
         $matchingGPOsArray += $gpo.DisplayName
      }
   }
}

# Define the list of OUs
$OUs = @(
   "OU=DoD WinSvr 2019 MS,DC=blackcloudnetwork,DC=local",
   "OU=DoD WinSvr 2019 DC,DC=blackcloudnetwork,DC=local",
   "OU=DoD WinSvr 2022 MS,DC=blackcloudnetwork,DC=local",
   "OU=DoD WinSvr 2022 DC,DC=blackcloudnetwork,DC=local",
   "OU=DoD Windows 10,DC=blackcloudnetwork,DC=local",
   "OU=DoD Windows 11,DC=blackcloudnetwork,DC=local",
   "OU=DOD Service Accounts,DC=blackcloudnetwork,DC=local",
```

```
    "OU=DOD Admins,DC=blackcloudnetwork,DC=local",
    "OU=DOD Users,DC=blackcloudnetwork,DC=local",
    "OU=DOD Groups,DC=blackcloudnetwork,DC=local"
)

# Link matching GPOs to each OU
foreach ($OU in $OUs) {
    foreach ($GPO in $matchingGPOsArray) {
        New-GPLink -Name $GPO -Target $OU
        Write-Host "Linked GPO '$GPO' to OU '$OU'"
    }
}
```

4 – GPO Exceptions

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```
# Import the Active Directory module
Import-Module ActiveDirectory

# Get all OU names
$OUs = Get-ADOrganizationalUnit -Filter * | Select-Object -ExpandProperty Name

# Loop through each OU
foreach ($OU in $OUs) {
    # Create GPO based on OU name
    $GPOName = "$OU - Exception"
    New-GPO -Name $GPOName -Comment "Created based on OU $OU"

    # Link GPO to the corresponding OU
    $OUPath = Get-ADOrganizationalUnit -Filter {Name -eq $OU} | Select-Object -ExpandProperty DistinguishedName
    New-GPLink -Name $GPOName -Target $OUPath | Set-GPLink -Order 1
}
```

5 - GPO Link OS

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```
# Define target OUs
$DoDWindows10Target = "OU=DoD Windows 10,DC=blackcloudnetwork,DC=local"
$DoDWindows11Target = "OU=DoD Windows 11,DC=blackcloudnetwork,DC=local"
$DoDWinSvr2019DCTarget = "OU=DoD WinSvr 2019 DC,DC=blackcloudnetwork,DC=local"
$DoDWinSvr2019MSTarget = "OU=DoD WinSvr 2019 MS,DC=blackcloudnetwork,DC=local"
$DoDWinSvr2022MSTarget = "OU=DoD WinSvr 2022 MS,DC=blackcloudnetwork,DC=local"
$DoDWinSvr2022DCTarget = "OU=DoD WinSvr 2022 DC,DC=blackcloudnetwork,DC=local"

# Read the CSV file
$csvData = Import-Csv -Path "C:\Oct 23 DISA STIG GPO Package 1001\Support Files
\DISA_AllGPO_Import_Oct2023.csv"

# Loop through each row in the CSV
foreach ($row in $csvData) {
   # Match the GPO name and assign the appropriate target
   switch -Wildcard ($row.GPOName) {
      "*DoD Windows 10*" {
         $target = $DoDWindows10Target
      }
      "*DoD Windows 11*" {
         $target = $DoDWindows11Target
      }
      "*DoD WinSvr 2019 DC*" {
         $target = $DoDWinSvr2019DCTarget
      }
      "*DoD WinSvr 2019 MS*" {
         $target = $DoDWinSvr2019MSTarget
      }
      "*DoD WinSvr 2022 MS*" {
         $target = $DoDWinSvr2022MSTarget
      }
      "*DoD WinSvr 2022 DC*" {
         $target = $DoDWinSvr2022DCTarget
      }
```

```
        default {
            Write-Host -ForegroundColor Red "No matching target found for $($row.GPOName)"
            continue
        }
    }

    # Generate the new-gplink command
    $newGplinkCommand = "new-gplink -name `"$($row.GPOName)`" -Target ""$target"" | Set-GPLink -Order 2"

    # Execute the command
    Invoke-Expression $newGplinkCommand
}
```

7 Book.csv

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DoD WinSvr 2019 MS
DoD WinSvr 2019 DC
DoD WinSvr 2022 MS
DoD WinSvr 2022 DC
DoD Windows 10
DoD Windows 11
DOD Service Accounts
DOD Admins
DOD Users
DOD Groups

Add to **A** Column and save a book.csv in c:\it\

2 - GPO import

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```
$importgpofile = Get-ChildItem -Path "C:\it\" -Filter 'DISA_All*'
$source = 'C:\it\'
$dest = 'C:\temp\'


##Search for STIG file and export
## Get-ChildItem -Path $source -Filter *STIG*.zip | Expand-Archive -DestinationPath $dest\STIGs\ -Force
###

Get-ChildItem -Path $source -Filter *STIG*.zip | sort lastwritetime | select -Last 1 | Expand-Archive -DestinationPath 'C:\temp\STIGs\' -Force

## Don't need this   $Foldername = Get-ChildItem -Path C:\temp\ -Filter "*DISA STIG GPO*"

##Setting Envirnoment Variable to copy file to later
$myfolder = [Environment]::GetFolderPath("MyDocuments")
$disaImportscript = Join-Path -Path $dest -ChildPath "STIGs\Support Files\DISA_GPO_Baseline_Import.ps1"

#Copy Migration table file to Mydocuments
Copy-Item -path "$source\importtable.migtable" -Destination $myfolder

##Testing
Copy-Item -path "$source\blackcloudNetwork-Sample.migtable" -Destination $myfolder

##Setting Varialbe to modify CVS

# Search for files named like "DISA_All" in C:\temp\ and its subdirectories
$fullyQualifiedPaths = (Get-ChildItem -Path "C:\temp\STIGs" -Filter "DISA_All*" -Recurse -File | sort lastwritetime | select -Last 1 ).FullName

# Display the fully qualified paths stored in the variable $fullyQualifiedPaths
Write-Host "Fully Qualified Paths:"
$fullyQualifiedPaths


  Get-Content $fullyQualifiedPaths |
  Where-Object { -not ($_.Contains('2013') -or $_.Contains('2016') -or $_.Contains('2012')) } |
```

```
    Set-Content 'C:\temp\STIGs\Support Files\DISA_AllGPO_Import_no201320162012.csv'



#This is needed because they had a type O in folder.  Discovered in the Q1 2024 STIG Release.
Get-ChildItem -Path "C:\temp\STIGs" -Directory -Filter "DoD Internet Explorer 11 v2r5" | ForEach-Object
{
    $newName = $_.Name -replace '^DoD Internet Explorer 11 v2r5', 'DoD Internet Explorer 11 v2r4'
    Rename-Item -Path $_.FullName -NewName $newName
}



Copy-Item 'C:\temp\STIGs\Support Files\DISA_AllGPO_Import_no201320162012.csv' -Destination
$myfolder


$sourcePath = "C:\temp\STIGs"
$destinationPath = "C:\Oct 23 DISA STIG GPO Package 1001"

Copy-Item -Path $sourcePath -Destination $destinationPath -Recurse -Force -ErrorAction
SilentlyContinue



&$disaImportscript


Write-Host -ForegroundColor Green " Select the CSV file First, Then select the MIG Table second"


Write-Host -ForegroundColor Green "Linking COMMON GPO to the Organization Units, space to
continue"
```